



WEI WANG

Computer Architect with 20+ years of experience bridging silicon engineering and compute infrastructure build-out. Currently leading CPU strategy for ByteDance, overseeing roadmap alignment with major silicon vendors to support massive-scale AI and general compute growth. Proven ability to translate deep technical analysis into high-impact executive strategies that optimize TCO and performance for next-generation AI workloads.

EDUCATION

Master of Philosophy | Engineering

University of Cambridge

Sept. 2004 – May 2006

Cambridge, England

Bachelor of Engineering | Electrical Engineering

Tongji University

Sept. 1999 – July 2003

Shanghai, China

WORK EXPERIENCE

Computer Architect

ByteDance Inc.

August 2023 – Current

San Jose, California

- **Global CPU Strategy:** Lead CPU product roadmap for compute infrastructure supporting 1M+ annual server growth. Define the "Golden SKU" architecture for AI and compute shapes.
- **Strategic Vendor Management:** Drive long-term silicon roadmaps with executive architects at NVIDIA, Arm, Qualcomm, Intel, and AMD. Secured critical architectural customizations (e.g., Vera, GNR, Turin) to align future hardware with internal AI/ML workload requirements.
- **TCO & Performance Optimization:** Achieved >15% Perf/TCO improvement G2G by optimizing the hardware stack for specific internal workloads (compute, database, storage, AI/ML).
- **AI Infrastructure Innovation:** Spearheaded proof-of-concepts for next-gen AI memory systems, including LPDDR-PIM and CXL-based expansion.
- **Cost Management:** Spearheaded corporate efforts in DRAM cost management in AI infra buildout. Explored DRAM hedging strategies to neutralize DRAM price fluctuations.
- **Team Management:** Ran virtual teams of 10+ members, driving workloads analysis and CPU customization.

Computer Architect

Arm Ltd

April 2006 – August 2023

Cambridge, England

- **Strategic ISA & IP Delivery:** Directed systems research on non-volatile memory (NVM), delivering 12+ patented IPs that directly influenced Arm's persistence support across industry protocols including CXL, AMBA CHI, and PCIe.
- **Technology Commercialization:** Successfully transferred architectural innovations to Arm's Chief Architect for ISA adoption, including IP US10,642,743, which was licensed to the JEDEC DDR4/DDR5 NVDIMM standards.
- **Executive Business Case Development:** Partnered with Arm's New Business Ventures to lead M&A technical due diligence, developing business cases with market sizing and ROI analysis for presentation to the Arm Board of Directors.
- **Accelerator Strategy:** Led the investigation into spatial architectures for general-purpose acceleration, achieving a 10x performance density increase for ML and 5G infrastructure workloads through hardware-software co-design.
- **Cross-Functional Leadership:** Managed a global team of 10+ engineers and co-supervised PhD research, while leading Arm's participation in multi-million dollar international consortia like EU Horizon 2020 SAGE2.
- **Ecosystem Orchestration:** Spearheaded the adoption of Arm servers in the healthcare and biotech sectors, optimizing genomics pipelines and integrating bit-shuffle instructions into the SVE2 (Armv9) ISA.

Architectural support for cloud accelerators | *Workloads, Architecture* 2022-2023

- **Explored architectural support for cloud accelerators tightly coupled with Arm CPUs**, in collaboration with hyperscaler partners to define use cases and requirements.
- Surveyed the accelerator design space and interfaces; developed a taxonomy to guide architectural integration strategies.
- Proposed architectural support for asynchronous data movement acceleration and opportunities in data center tax workloads (e.g., Protobuf, hashmap).
- Investigated architectural extensions to accelerate dynamic programming patterns in cloud environments.

Spatial architecture | *Architecture, μ Arch, Workloads* 2021-2022

- **Explored spatial architecture for general-purpose acceleration**, targeting **10 $\times$ performance density** over CPUs with Scalable Matrix Extension.
- Developed and evaluated a spatial dataflow ISA with features like decoupled access/execute, reconfigurable distributed memory, and sparsity support across many-PE architectures.
- Achieved 10 $\times$ performance density on infrastructure workloads (e.g., 5G, ML) through handcrafted kernels (e.g., GEMM, Cholesky), leveraging tiling, vectorization, and pipelining in co-designed triggered assembly.
- Collaborated on ISA development, simulator extension, architectural design space exploration, and HW implementation for performance density benchmarking.

Architectural support for persistent memory | *Architecture, Microarchitecture* 2018-2021

- **Pioneered architectural support for a spectrum of memory persistency models**, from relaxed (release, strand) to strict (sequential, strict persistency), with full-stack contributions spanning programming languages, ISA, system architecture, and microarchitecture.
- Published 12+ papers at top venues (PLDI'18, ISCA'20, HPCA'21); **IEEE Micro Top Picks 2019**; invited keynotes at PIRL, VCEW, Dagstuhl, and NANDA.
- Delivered 12+ patented IPs with lasting industry impact; influenced Arm's persistence support across AMBA CHI and industry protocols (CXL, CCIX, PCIe, Gen-Z, NVDIMM-P).
- IP **US10,642,743** licensed to JEDEC DDR4/DDR5 NVDIMM standard, driving standards-level adoption of persistency innovations.

Exploring NVM as last-level cache | *Microarchitecture* 2019

- **Explored on-chip use of NVM** within application processor cache hierarchies, MCUs, and accelerator memory subsystems.
- Delivered architecture-to-silicon analysis with quantified trade-offs on power, performance, area, endurance, and retention.
- Influenced Arm's physical IP and CPU design divisions with design recommendations for NVM integration.
- **Led a cross-functional global team of four engineers** spanning architecture and silicon implementation.

Addressing the endurance challenge of NVM | *Microarchitecture, Architecture, Software* 2015-2017

- **Proposed and evaluated software-managed wear-levelling and write-reduction techniques** for NVM, including compression between LLC and NVM, cache replacement policies to retain dirty data longer, and architectural support for dead-write elimination.
- Results informed CeRAM/MRAM teams at Arm and were presented at **FAST'19** and **MEMSYS'17** conferences.

Pushing Arm servers into healthcare | *Software, Architecture* 2016-2017

- **Led Arm's ecosystem efforts to drive adoption of Arm servers in healthcare and biotech sectors**, collaborating across marketing, engineering, server vendors, and pharmaceutical customers.

- Optimized genomics pipelines on Arm: enhanced glibc functions, accelerated bioinformatics kernels (Banded Smith-Waterman, HMM) using Arm SIMD and partner SoC accelerators, and improved auto-vectorization via Arm HPC Compiler with NEON/SVE; showcased at Arm Annual Partners Meeting 2017.
- Contributed to scalable vector ISA extensions for genomics, including bit shuffle instructions integrated into SVE2 (Armv9).
- Partnered with Arm's New Business Ventures on **acquisitions**, developing business cases with market sizing, budgeting, recruiting, and ROI analysis presented to the Arm Board.

Data profiling tool for exposing data movements in memory | *Software, Microarchitecture* 2013-2015

- **Developed a C++ data profiling tool based on gem5 to analyze CPU cache interactions and optimize data movement for improved performance and energy efficiency.** Enabled developers to identify hot data structures and optimize access patterns, reducing cache misses.
- Presented at Arm Global Engineering Conference 2014; won Best Poster Award at MSPC 2014.
- Prototype integrated into Arm's flagship performance analysis tool, DS-5 Streamline.

Android Dalvik virtual machine performance analysis and optimizations | *Runtime, Compilers* 2012-2013

- **Conducted competitive performance analysis of the Dalvik virtual machine on Arm**, examining the Android Dalvik C++ codebase to identify gaps versus Java SE Embedded C1/C2 implementations.
- Delivered actionable recommendations for Dalvik JIT compiler optimizations to improve runtime performance.

gem5 simulator development & mobile systems research | *Simulator, Workloads & μ Arch* 2010-2012

- **Extended gem5 open-source full-system simulator to enhance support for Arm CPUs and systems**, developing C++ modules for Arm GDB client, VNC (X11) client, Armv8 architecture verification and bug fixes, Linux kernel bring-up, memory system restructuring, performance optimizations, and Android workload image creation.
- Enabled realistic Android workloads simulation on Armv8, uncovering microarchitectural insights such as large instruction footprints and instruction cache miss reduction needs beyond conventional benchmarks.
- Presented findings at IISWC'13, MEMSYS'15, and published on arXiv'20.

Coarse-grained reconfigurable architecture | *Workloads & Design Space Exploration* 2008-2010

- **Led workload development and design space exploration for a coarse-grained reconfigurable architecture (CGRA) aimed at competing with bit-configurable FPGAs on PPA (performance, power, area).**
- Created a hardware benchmark suite of 25 diverse applications—from quantitative finance (Black-Scholes) to wireless communications (FFT, Viterbi)—implemented in Verilog and C for hardware and CPU evaluation.
- Enabled automated comparison of CGRA designs against CPU, FPGA, and ASIC platforms, providing critical insights into architectural trade-offs across varied workloads.

Razor: dynamic voltage scaling based on timing speculation | *SoC Prototyping & Verification* 2006-2008

- **Advanced the Razor low-power voltage-margin shaving technology (2021 Micro Test of Time award winner) at Arm Research.**
- Designed and prototyped a Razor SoC based on ARM1176JZF-S, implemented on FPGA; developed low-level C and ARM assembly software for verification and benchmarking.
- Built and brought up Linux kernel and workloads on the platform, demonstrating Razor's viability for Arm CPUs.
- Successfully transferred the technology to Arm CPU product teams for integration.

PUBLICATIONS AND PRESENTATIONS

Intel Arch Forum Jun'25 *Invited Talk*: ByteDance CPU Customization
NANDA Workshop 2022 *Invited Talk*: Architectural Support for Persistent Memory
TCAD'22 Pragmatic Memory-System Support for Intermittent Computing using Emerging Non-Volatile Memory
Dagstuhl Seminar 21462 *Invited Talk*: Architectural Support for Persistent Programming
VCEW'21 *Invited Talk*: Architectural Support for Persistent Memory
HPCA'21 BBB: Simplifying Persistent Programming Using Battery-Backed Buffers
PIRL'20 *Invited Keynote*: Architectural Support for Persistent Programming
arXiv'20 The gem5 Simulator: Version 20.0+
ISCA'20 Relaxed Persist Ordering Using Strand Persistency
ISPASS'20 Fused: Closed-loop Performance and Energy Simulation of Embedded Systems
SPAA'19 Persistent Atomics for Implementing Durable Lock-Free Data Structures for Non-Volatile Memory
IEEE Micro Top Picks 2019 Language Support for Memory Persistency
FAST'19 Software Wear Management for Persistent Memories
DAC'19 Efficient State Retention Through Paged Memory Management for Reactive Transient Computing
PLDI'18 Persistency for Synchronization-Free Regions
MEMSYS'18 Quantifying the Performance Overheads of PMDK
MEMSYS'17 Composing Lifetime Enhancing Techniques for Non-Volatile Main Memories
MEMSYS'15 Inefficiencies in the Cache Hierarchy: A Sensitivity Study of Cacheline Size with Mobile Workloads
IISWC'13 A Structured Approach to the Simulation, Analysis and Characterization of Smartphone Applications

PATENTS

WO2025210327 Vector Combination Instruction
UK2218617.5 Triggered-producer and Triggered-consumer Instructions
US17,942,554 Issuing a Sequence of Instructions Including a Condition-Dependent Instruction
UK2105240.2 Apparatus and Method for Generating Debug Information
US11,740,908 Systems and Methods for Defining and Enforcing Ordered Constraints
US11,513,962 Draining Operation to Cause Store Data to be Written to Persistent Memory
GB2,598,784 Draining Operation for Draining Dirty Cache Lines to Persistent Memory
US11,409,617 Hardware Bitvector Techniques
GB2,591,813 Task-Aware Checkpointing for Intermittent Compute
US10,956,166 Instruction Ordering
US10,866,890 Method and Apparatus for Implementing Lock-Free Data Structures
US10,445,238 Robust Transactional Memory
US11,182,106 Energy Conservation for Memory Applications
US10,831,678 Multi-Tier Cache Placement Mechanism
US11,137,919 Initialisation of a Storage Device
US10,642,743 Apparatus and Method of Handling Caching of Persistent Data
US10,964,386 Initialisation of a Storage Device
US10,847,204 Control of Refresh Operation for Memory Regions
US10,719,236 Non-Volatile Buffer for Memory Operations

HONORS AND AWARDS

- Language Support for Memory Persistency was selected as IEEE Micro Top Picks 2019** May 2019
Recognition for the top 12 most influential publications of the year in computer architecture
- Best Poster Award at MSPC 2014** June 2014
DataProf: Exposing Data Movements Across the Memory Hierarchy
- Arm Multiple Patent Grant Award** 2020, 2021, 2022
Awarded for every fourth, tenth and twenty-fifth granted US patents

SERVICES AND COMMUNITY

- Organized CPU technical workshops with Intel (Q4'24/25), AMD (Q4'24/25), and Arm (Q1'25/Q4'25) at ByteDance (San Jose)
- Member of HiPEAC, Member of SNIA NVM Programming TWG
- Technical Program Committee of HPCA'20, NVMW'20, NVMW'21, NVMW'23, MEMSYS'23, VCEW'25
- Organizing Committee of IISWC'20
- Organizing Committee, Technical Program Committee and Session Chair for Arm Research Summit 2016-2020
- Reviewer for IEEE Computer Architecture Letters (CAL), IEEE Transactions on Emerging Topics in Computing (TETC), and ACM Transactions on Architecture and Code Optimization (TACO)
- Organizer of Architectural Exploration with Gem5 Tutorial at ASPLOS 2017
- Member of Arm Patents Review Committee (PRC)